

over time, the numerator and denominator will be the same, so the VR will be 1. In a mean-reverting market, returns are negatively correlated, and the VR will be less than 1. In a trending market, returns will be positively correlated, and the VR will be greater than 1. If we calculate the VR over various k periods, we will be able to tell if the market has been trending or mean reverting over those periods.

Tony Cooper provided the charts for [Figure 9.1](#).³ [Figure 9.1a](#) shows the S&P 500 identified by 15-year time intervals. [Figure 9.1b](#) shows the VR associated with each of these 15-year intervals. We see that for the 1999–2013 interval (the lowest line on the VR chart), the market was mean-reverting over all the k periods. For other 15-year intervals, the VR values are mostly higher than 1, indicating trending markets. The diverging VR chart patterns show us that backtests performed on only 15 years of data are unlikely to give reliable forward-looking predictions. One of the most frequent mistakes I see others make is using only a limited amount of data, such as 15 years, to develop a model and then expect it to give reliable results going forward in time. The prominent researcher Kenneth French (Fama and French, 2007) once said that 78 years of data in the Center for Research in Security Prices (CRSP) database might not be enough to separate out noise from one's results.⁴ We saw in [Chapter 6](#) that respected researchers determined there was a significant value premium based on 28 years of data. Now that substantially more data is available, we see that the value premium, like the size premium before it, might be suspect.